

Soumajit Roy

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Commissioning Engineer with hands-on experience in KAVACH (TCAS) deployment, testing, and commissioning for Indian Railways. Experienced in FAT, I-SAT, SAT, locomotive trials, RFID verification, troubleshooting, and system validation. Successfully contributed to commissioning of 100 km of KAVACH territory under Central Railway and possess a strong foundation in railway signalling and train protection systems.

Skills

Railway Signalling & Kavach: Kavach (TCAS), ATP, ETCS, ERTMS, Railway Signalling Systems, Interlocking Principles

Commissioning & Testing: FAT, I-SAT, SAT, Loco Trials, RFID Tag Verification, System Testing, Troubleshooting

Tools & Documentation: AutoCAD 2D, AutoCAD Electrical, SIP, Control Tables, Technical Documentation, IREPS

Experience

Commissioning Engineer, HBL Engineering Limited – Solapur Division (KAVACH Project) Dec 2025– Present

- Commissioned approximately **100 km of KAVACH territory between Daund and Kurduwadi**, supporting successful deployment and operational readiness of the system.
- Executed **Factory Acceptance Tests (FAT), Integrated Site Acceptance Tests (I-SAT), and Site Acceptance Tests (SAT)** for KAVACH systems, ensuring compliance with project specifications.
- Participated in **locomotive trials, functional testing, and system validation activities** for onboard and trackside KAVACH equipment.
- Supervised **RFID tag installation, mapping, and inter-tag distance verification**, ensuring accurate train positioning and reliable system performance.
- Performed pre-commissioning inspections, checklist validation, and system readiness verification prior to commissioning.
- Diagnosed and resolved commissioning-stage issues in onboard and trackside subsystems, minimizing project delays and improving system reliability.
- Coordinated with signalling, telecom, and project teams to ensure timely execution of commissioning activities.
- Prepared commissioning reports, test records, technical documentation, and conducted training sessions for railway personnel on KAVACH operation and basic troubleshooting.

Trainee Engineer (S&T), Railtranz Consultancy Pvt. Ltd. – New Delhi Aug 2024 – Nov 2025

- Contributed to signalling remodification projects under **West Central Railway and Western Railway**, supporting station modernization and infrastructure upgrades.
- Prepared signalling documents including **SIP, Control Tables, CTR, CCP, CRP, CCC, TBP, Location Details, and Power Supply Diagrams** in accordance with Indian Railways standards.
- Developed signalling layouts and technical drawings using **AutoCAD**, improving documentation accuracy and design efficiency.
- Supported **IREPS tender preparation and technical documentation** for railway signalling projects.
- Completed extensive training in railway signalling, interlocking principles, and KAVACH (TCAS), building a strong foundation in train protection systems.

Certifications

- KAVACH (TCAS) Training – Narayan Rail Academy (2025)
- AutoCAD Electrical 2024
- IRISSET Railway Signalling Learning Modules
- PLC Automation Certification
- Python Programming Certification

Project Portfolio

Portfolio: soumajitroy.com

- Railway signalling drawings and documentation
- Kavach (TCAS) learning and technical content
- Professional engineering projects and certifications

Education

NSHM Knowledge Campus, Durgapur

2020 – 2023

Bachelor of Technology (B.Tech), Electrical Engineering

CGPA: 9.02 / 10

New Horizons Institute of Technology, Durgapur

2017 – 2020

Diploma in Electrical Engineering

CGPA: 7.60 / 10